

## NOTES

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BENTHIC MACROINVERTEBRATE ASSEMBLAGE STRUCTURE IN  
SALINIZED REACHES OF THE LOWER PECOS RIVER, TEXAS

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**ABSTRACT**—In the Permian Basin region (New Mexico and Texas), salinization has occurred in the Lower Pecos River as natural saline inputs have been exacerbated by flow-regime change and other anthropogenic effects. In this study, we deployed Hester-Dendy multi-plate samplers for two seasons to assess the structure of colonizing benthic macroinvertebrate assemblages at three sites in the Lower Pecos River (two in the salinized Permian Basin region, one in the transition to the Edwards Plateau). Taxonomic richness was low, with four to nine families colonizing samplers. Highest richness was observed at our study site near Orla, Texas, during a period of higher discharge. Assemblages were dominated by salt-tolerant taxa, and groups considered particularly sensitive to salinization (Ephemeroptera, Plecoptera, and Trichoptera) were absent except for two tolerant trichopteran families. Because river salinization is expected to increase with climate change, understanding its effects on the structure of biotic assemblages is important for conservation.

**RESUMEN**—En la región de la Permian Basin (Nuevo México y Texas), la exacerbación de sales de origen natural como resultado de cambios en el régimen de caudal y otros impactos antropogénicos han llevado a la salinización del río Pecos. En este estudio, utilizamos muestreadores de placas múltiples Hester-Dendy durante dos temporadas para evaluar la estructura de las comunidades de macroinvertebrados bentónicos colonizadores en tres sitios en el río Pecos (dos en la región salinizada de la Permian Basin, uno en la transición a la Edwards Plateau). La riqueza taxonómica fue baja, con cuatro a nueve familias colonizando los muestreadores. Observamos la mayor riqueza en el sitio cerca de Orla, Texas, durante un período de mayor caudal. Las comunidades estuvieron dominadas por taxones tolerantes a la salinidad y los grupos considerados particularmente sensibles a la salinización (Ephemeroptera, Plecoptera y Trichoptera) estuvieron ausentes, excepto por dos familias de tricópteros tolerantes. Debido a que se espera que aumente la salinización de los ríos con el cambio climático, es importante para la conservación comprender sus efectos en la estructura de comunidades bióticas.

Many dryland rivers in the southwestern United States are experiencing anthropogenic salinization associated with irrigation practices, high rates of evaporation in reservoirs, and reduced fresh discharge due to dams, diversions, and groundwater over-extraction (e.g., Carlson and Muth, 1989; Hoagstrom, 2009). In these systems, the natural salt balance has been disrupted as salts have accumulated (e.g., through irrigation return flows), and diluting flows—such as seasonal flood pulses—are no longer adequate to lower salt concentrations. Climate change and increasing water demand are expected to intensify this situation (Nielsen and Brock, 2009; Cañedo-Argüelles et al.,

2013). The effects of salinization on the structure and function of dryland river ecosystems are not completely understood, but increasing salt concentrations are known to often have strong effects on freshwater organisms (reviewed by Hart et al., 1991; Cañedo-Argüelles et al., 2013). For benthic macroinvertebrates in rivers, salinization is expected to diminish diversity in local assemblages as salt-intolerant species are excluded (Pinder et al., 2005; Kefford et al., 2016). This often disproportionately affects certain functional groups, such as shredders that process leaf litter and other coarse particulate organic matter, potentially disrupting organic

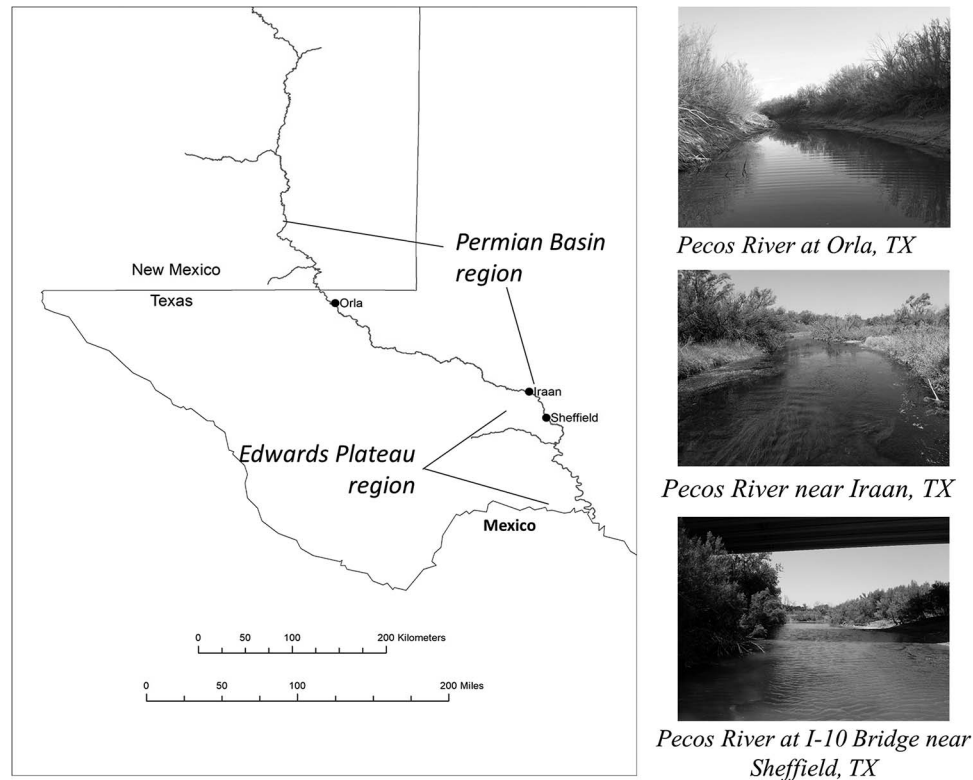


FIG. 1—Map of study sites on the Pecos River in New Mexico and Texas, USA. Dots indicate locations of study sites (2017) where Hester-Dendy artificial substrates were deployed. The two geomorphic regions of the lower Pecos basin (Permian Basin and Edwards Plateau) are indicated.

matter processing and energy flow in river food webs (Bailey et al., 2006; Schäfer et al., 2012; Boersma and Lytle, 2021).

In the Permian Basin region of Texas and New Mexico, anthropogenic salinization in the Pecos River has been caused largely by flow alteration and irrigation practices (Hoagstrom, 2009). Although brine springs contribute salts to the Pecos, these inputs were a small fraction of natural river flow prior to development, and the river historically supported a more diverse freshwater fauna (Davis, 1987; Linam and Kleinsasser, 1996; Hoagstrom, 2003). Diminished spring inputs since aquifer depletion began around the turn of the 20th Century, and the dampening of flow pulses by a series of dams, have left the Lower Pecos River in the Permian Basin with insufficient fresh discharge for salt export (Dearen, 1996; Hoagstrom, 2009). It is now common to record specific conductivity levels above 30,000  $\mu\text{S}/\text{cm}$  (19.2 ppt) in reaches below Red Bluff dam in the Permian basin (East et al., 2017; Delaune, 2020). Fish diversity has severely declined in the salinized Lower Pecos River in the last several decades (Hoagstrom, 2003; Cheek and Taylor, 2016), but less is known about how salinization has affected benthic macroinvertebrate assemblages.

In the 1970s, Davis (1980) carried out an extensive study of benthic macroinvertebrate diversity in the Lower Pecos River. He found that higher salinity in the Permian

Basin depressed macroinvertebrate richness and that salinized habitats contained several coastal saltwater taxa. However, effects on benthic macroinvertebrate assemblages have not been examined in recent decades as salinization has intensified. In this study, our objective was to assess the structure of benthic macroinvertebrate assemblages colonizing artificial substrates at three sites in the Lower Pecos River. As outlined in the conceptual framework of Kefford et al. (2016), we expected salt-intolerant insect taxa such as Ephemeroptera, Plecoptera, and Trichoptera would be absent in sites with high specific conductivity, whereas more salt-tolerant taxa such as crustaceans and certain families of Odonata, Coleoptera, and Diptera would predominate. We predicted that across sites, we would see evidence of diminished diversity in macroinvertebrate assemblages approximately 40 years after the surveys of Davis (1980).

Our study sites were in the main stem of the Lower Pecos River near Orla, Iraan, and Sheffield, Texas (Fig. 1). The sites at Orla and Iraan were located within the Permian Basin region below Red Bluff dam, where discharge is typically lowest and conductivity is highest. The Orla study site (at the F.M. 652 crossing) was the same site sampled by Davis (1980), and our Sheffield site (at the I-10 crossing) was approximately 10 river kilometers (rkm) above the site sampled in the previous study. Our Iraan study site (at the S.H. 349 crossing) was not

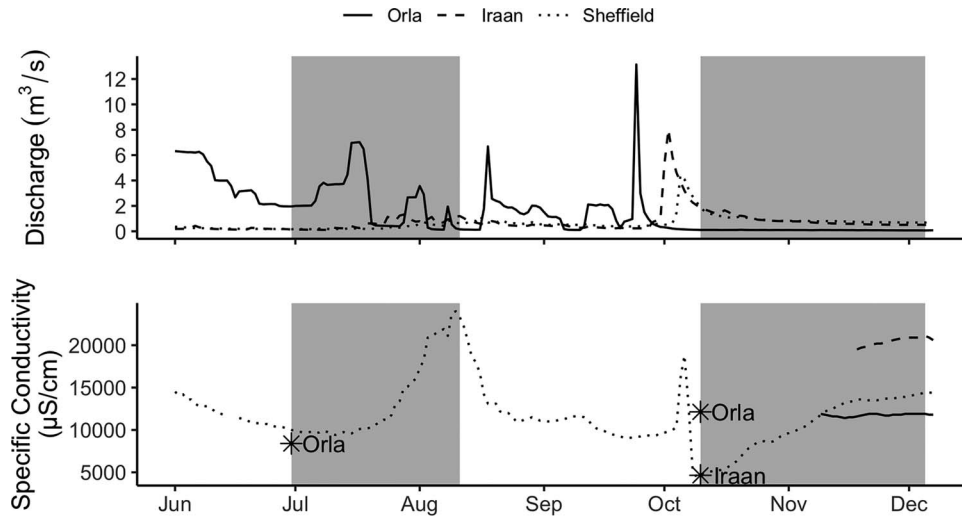


FIG. 2—Hydrograph of river discharge during our summer and fall 2017 sampling periods (shaded sections). Discharge data are from United States Geological Survey (USGS) gauge 08412500 at the Orla site, USGS gauge 08446500 approximately 55 river kilometers (rkm) above the Iraan site, and USGS gauge 08447000 approximately 9 rkm below the Sheffield site. Continuous conductivity data (lower panel) were available at the Sheffield gage for our entire study period, but only from mid-November to December for the gages at Orla and above Iraan. For those two sites, our instantaneous measures of specific conductivity (asterisks) are included (the Iraan site was not sampled in summer).

included in the Davis (1980) study. The Sheffield site is below the Permian Basin region in the transition to the Edwards Plateau region, but above where inputs from Independence Creek dilute salt concentrations, so conductivity typically remains high (Linam and Kleinsasser, 1996). During our study, the Orla study site had mostly mud and silt substrate, a predominance of slow-run habitat types, and abundant widgeon grass (*Ruppia maritima*). At the Iraan site, the charophyte *Chara* was abundant and substrate was composed of mostly mud and silt with small patches of gravel. Small riffles were available along with slow, shallow runs at this site. At the Sheffield site, the river substrate was composed of boulders, cobble, gravel, sand, and mud with very few aquatic macrophytes. Multiple habitat types including runs, riffles, and pools were present at the Sheffield site.

We sampled during summer (July–August) and fall (October–December) of 2017. In comparison to Davis (1980), our study did not include winter and spring samples. At each site we measured habitat variables including flow velocity, specific conductivity, temperature, and dissolved oxygen. We retrieved daily discharge and conductivity data for the study period from three United States Geological Survey (USGS) flow gages: USGS 08412500 at the Orla site, USGS 08446500 approximately 55 rkm above the Iraan site, and USGS 08447000 approximately 9 rkm below the Sheffield site. At each site and season, we deployed five Hester-Dendy multi-plate samplers (NKY Environmental Supply, Union, Kentucky) attached to a cinderblock. The samplers consisted of eight square hardboard plates ( $7.6 \times 7.6 \times 0.3$  cm) with spacers ( $2.5 \times 2.5 \times 0.3$  cm) arranged to form three single spaces, three double spaces, and one triple space between the plates

(Ohio Environmental Protection Agency [EPA], 2015). The samplers provided standardized substrate areas with physically consistent hard surfaces, allowing us to assess the influence of water quality on the benthic macroinvertebrate assemblage structure (Richards et al., 1993; Ohio EPA, 2015). We placed samplers in shallow run habitats approximately 1 m from shore in water depths of 30–100 cm and flow velocities of 0.1–0.4 m/s. We left each sampler in the field for 6 weeks in the summer and 8 weeks in fall. After this time, we collected the samplers and stored them individually in bags with 70% ethanol. In the laboratory, we scraped all macroinvertebrates from the plates and sorted them by taxonomic family. Samplers were lost during the summer survey at the Iraan site, so we only report fall results for that site.

In summer, discharge was higher at the Orla site (up to  $7.1 \text{ m}^3/\text{s}$ ) compared with the Sheffield site ( $0.1\text{--}1.2 \text{ m}^3/\text{s}$ ; Fig. 2). Correspondingly, conductivity was lower in the Orla site ( $8,403 \mu\text{S}/\text{cm}$ ) compared with the Sheffield site ( $14,065 \mu\text{S}/\text{cm}$ ). The Iraan site had the highest recorded conductivity, with a value of  $24,937 \mu\text{S}/\text{cm}$  in June 2017. In fall, discharge was extremely low ( $\leq 0.1 \text{ m}^3/\text{s}$ ) at the Orla site and conductivity was  $11,690 \mu\text{S}/\text{cm}$ . At Iraan, discharge ranged from  $0.5$  to  $1.5 \text{ m}^3/\text{s}$  during fall sampling, and conductivity was relatively low ( $4,648 \mu\text{S}/\text{cm}$ ) when samplers were deployed following a flow pulse in early October (Fig. 2). Conductivity at Sheffield in fall was also lower ( $4,354 \mu\text{S}/\text{cm}$ ), with a discharge of  $0.7\text{--}1.3 \text{ m}^3/\text{s}$  during fall sampling.

Taxonomic richness (number of benthic macroinvertebrate families) was nine and six for the Orla site in summer and fall, respectively (Table 1). At the Iraan site (only sampled in fall), we collected five macroinvertebrate families on

TABLE 1—Abundance of benthic macroinvertebrates by site and season in the lower Pecos River in Texas during surveys in 2017. Functional feeding group assignments for taxonomic families were based on classification of most taxa within each family according to Merritt and Cummins (1996).

| Order       | Family          | Functional feeding group | Orla   |       | Iraan<br>Fall | Sheffield |      |
|-------------|-----------------|--------------------------|--------|-------|---------------|-----------|------|
|             |                 |                          | Summer | Fall  |               | Summer    | Fall |
| Coleoptera  | Gyrinidae       | Predator                 | —      | —     | —             | —         | 4    |
|             | Hydrophilidae   | Collector-gatherer       | 1      | —     | 3             | 2         | 1    |
| Diptera     | Ceratopogonidae | Predator                 | 3      | —     | —             | —         | —    |
|             | Chironomidae    | Collector-filterer       | 164    | 4,630 | 65            | 509       | 802  |
| Odonata     | Calopterygidae  | Predator                 | 2      | 2     | —             | —         | —    |
|             | Coenagrionidae  | Predator                 | 43     | 64    | —             | 68        | 49   |
|             | Gomphidae       | Predator                 | —      | 3     | —             | —         | —    |
| Trichoptera | Hydropsychidae  | Collector-filterer       | 15     | 9     | 3             | —         | —    |
|             | Hydroptilidae   | Scraper                  | 2      | 532   | —             | —         | 105  |
| Amphipoda   | Hyalellidae     | Collector-gatherer       | —      | —     | 671           | 6         | 5    |
| Gastropoda  | Physidae        | Scraper                  | 9      | —     | 209           | —         | 6    |
|             | Planorbidae     | Scraper                  | 5      | —     | —             | —         | —    |

samplers. At Sheffield, richness was four in the summer and increased to seven families in fall. At Orla and Sheffield, nonbiting midges (Diptera: Chironomidae) were the most abundant taxon collected in all samplers, followed by narrow-winged damselflies (Odonata: Coenagrionidae) in summer and microcaddisflies (Trichoptera: Hydroptilidae) in fall. Chironomids and hydroptilid caddis larvae occurred at very high densities in Orla samplers during the fall season. The Iraan site had a very distinct assemblage structure characterized by high densities of amphipods and gastropods.

As a result of high discharge in June and July 2017, conductivity was lower at the typically more salinized Orla site compared with the Sheffield site, which did not experience a diluting flow pulse of the same magnitude in that period. Relatively low salinity at the Orla site during higher-discharge conditions might have promoted greater diversity of benthic macroinvertebrates. In other stream systems, benthic macroinvertebrate richness has been shown to increase—with addition of more sensitive taxa—following moderate flow pulses (Stitz et al., 2016). The pattern changed in fall, when discharge decreased substantially and conductivity increased at Orla, and richness of benthic macroinvertebrates was correspondingly lower. This suggests that in the Lower Pecos River, temporal variation in discharge and conductivity is important in driving benthic macroinvertebrate assemblage structure, as has been shown in other dryland rivers (Boulton et al., 1992; Marshall et al., 2006).

In comparison to the study by Davis (1980) in the 1970s, we found similar benthic macroinvertebrate richness at the Orla site in summer 2017 when conductivity was lower compared with the fall. However, we found much-lower richness at our Sheffield site in both seasons compared to the prior surveys (Table 2), and lower richness at Orla and Iraan in fall 2017 during low discharge (and high conductivity). Some taxa collected by Davis

(1980) that were absent in our surveys are indicators of good habitat quality that have likely been excluded from assemblages due to diminished discharge and poor water quality. For example, prong-gilled mayflies (Ephemeroptera: Leptophlebiidae) were historically collected at Sheffield, but we did not collect this taxon—or any other ephemeropterans—in any of our sites. Ephemeropterans are generally intolerant of high salinity (Berner, 1975), and leptophlebiid abundance is considered an indicator of good habitat quality (Brasil et al., 2014). Davis (1980) noted that the leptophlebiid species collected in his surveys at Sheffield, *Neochoroterpes oklahoma*, is somewhat euryhaline, but our results suggest that present-day conditions might be beyond its tolerance level. Leptophlebiids are regularly encountered in the main-stem Lower Pecos above and below our study region at the confluences of the Black River and Independence Creek (Delaune, 2020) where fresh discharge dilutes salinity and otherwise improves habitat quality (Pease and Delaune, 2021).

Davis (1980) also collected larval pyralid moths (Lepidoptera: Pyralidae) and dobsonflies (Megaloptera: Corydalidae) at Sheffield (Table 2). Although sensitivity of aquatic pyralid larvae to habitat stressors is poorly studied, they have been shown to be somewhat intolerant to stream habitat alteration (Masese et al., 2009). Corydalids are also sensitive to degraded water quality (Barbour et al., 1999) and are less abundant in altered stream habitats (Wantzen, 2006). We did not collect these taxa in either season at Sheffield, further indicating that habitat quality has degraded since the earlier work in the region. Though our measures of specific conductance at Sheffield were within the range reported by Davis (1980), discharge at this site was higher during the 1970s surveys. Opportunities for dilution in the present-day Lower Pecos have declined steadily due to diminished discharge over the last several decades, which means that the



TABLE 2—Percent of total individuals (summed across seasons) for benthic macroinvertebrates collected from the Lower Pecos River near Orla and Sheffield, Texas, by Davis (1980) and in the present study (2017). Note that differences in collection methods between the two studies may have influenced relative abundance of taxa.

| Order         | Family                       | Orla         |               | Sheffield    |               |
|---------------|------------------------------|--------------|---------------|--------------|---------------|
|               |                              | Davis (1980) | Present study | Davis (1980) | Present study |
| Coleoptera    | Gyrinidae <sup>a</sup>       | —            | —             | —            | 0.3           |
|               | Hydrophilidae                | —            | <0.1          | 0.4          | 0.2           |
| Diptera       | Ceratopogonidae              | 0.1          | 0.1           | 0.1          | —             |
|               | Chironomidae                 | 44.2         | 87.4          | 51.3         | 84.2          |
|               | Tabanidae <sup>b</sup>       | —            | —             | 0.2          | —             |
| Ephemeroptera | Leptophlebiidae <sup>b</sup> | —            | —             | 7.6          | —             |
| Lepidoptera   | Pyralidae <sup>b</sup>       | —            | —             | 0.1          | —             |
| Megaloptera   | Corydalidae <sup>b</sup>     | —            | —             | <0.1         | —             |
| Odonata       | Calopterygidae <sup>a</sup>  | —            | 0.1           | —            | —             |
|               | Coenagrionidae               | 2.1          | 2.0           | 4.4          | 7.5           |
|               | Gomphidae                    | 0.8          | 0.1           | <0.1         | —             |
| Trichoptera   | Hydropsychidae               | 26.2         | 0.4           | 19.8         | —             |
|               | Hydroptilidae                | 7.3          | 9.7           | 8.7          | 6.7           |
| Amphipoda     | Hyalellidae                  | —            | —             | <0.1         | 0.7           |
| Gastropoda    | Physidae                     | —            | 0.2           | 1.5          | 0.4           |
|               | Planorbidae <sup>a</sup>     | —            | 0.1           | —            | —             |

<sup>a</sup> Family collected in present study but not by Davis (1980).

<sup>b</sup> Family collected by Davis (1980) but not collected in present study.

duration of poor water quality conditions has increased (Hoagstrom, 2009). Longer exposure to high salinity more thoroughly filters against macroinvertebrate taxa that require fresher conditions to complete their life cycles (Hassell et al., 2006; Kefford et al., 2007).

Other insects collected in our samplers were taxa known to tolerate salinization. We collected two caddisfly families, Hydropsychidae and Hydroptilidae, in our surveys of salinized Lower Pecos River sites. These were also the only trichopteran taxa collected in these sites by Davis (1980). Hydropsychids, a family of net-spinning caddisflies, have been documented in many high-salinity rivers (Gallardo-Mayenco et al., 1998; Piscart et al., 2005a), and some species are highly salt tolerant (Sala et al., 2016). Elsewhere in the southwestern United States, hydroptilid microcaddisflies are also abundant in benthic assemblages of salinized reaches of the lower Colorado River (Blinn and Ruiter, 2006). Water scavenger beetles (Coleoptera: Hydrophilidae) were also somewhat common in our samplers, and *Berosus*, the genus documented by Davis (1980), is considered salt tolerant (Kefford et al., 2006; Dunlop et al., 2008). Among odonates, Coenagrionidae was the most abundant taxon, and this family also withstands high salinity (Kefford et al., 2006; Dunlop et al., 2008). In agreement with the 1970s survey, we collected chironomids at all sites across seasons, though their relative abundance was higher in the present study (Table 2). Salinity tolerance varies widely among taxa in the family Chironomidae, but some genera are relatively salt tolerant (Bailey et al., 2006), including *Tanytarsus*,

which Davis (1980) documented as the most abundant chironomid genus in salinized Pecos sites. Biting midges (Diptera: Ceratopogonidae), collected in the historical survey and our study, are also generally considered tolerant of high conductivity (Kefford et al., 2006).

Notably, the benthic macroinvertebrate assemblage was quite distinct at the Iraan site, within the reach of the Lower Pecos where salinity is usually highest (Hoagstrom, 2009). Amphipods and physid snails were most abundant in samplers at that site, far outnumbering the only three insect families collected (Table 1). This is similar to what Bätke and Coring (2011) and Piscart et al. (2005b) observed in European salinized rivers, where benthic macroinvertebrate surveys revealed a shift from assemblages including more insect taxa toward assemblages dominated by amphipods and salt-tolerant mollusks following exposure to industrial salt pollution. The food-web effects of a shift from a macroinvertebrate assemblage comprising mostly insect larvae to one dominated by amphipods and other salt-tolerant, noninsect taxa (e.g., ostracods, salt-tolerant snails) are unknown and merit future research. Indeed, this is a pattern expected with river salinization because although insects typically dominate benthic macroinvertebrate assemblages in rivers, their evolutionary history constrains their ability to adapt to rising salinities (Kefford et al., 2016).

At all Lower Pecos River sites across both seasons, taxa belonging to the collector-gatherer and collector-filterer functional feeding groups dominated macroinvertebrate assemblages. Davis (1987) also concluded that macroinvertebrate

assemblages in salinized Pecos River reaches were characterized by tolerant, trophic generalists. Collectors as a group are considered tolerant of degraded habitats, and although they are typically common in dryland rivers (Bogan et al., 2013; Leiva et al., 2020), dominance by this group is considered a sign of diminished ecological integrity (Barbour et al., 1999). In other salinized rivers, reduced abundance of more-sensitive trophic specialists (e.g., shredders) and a shift toward dominance by collectors has been documented (Marshall and Bailey, 2004; Piscart et al., 2006). Reduced abundance of ecological specialists and somewhat functionally homogeneous assemblages are symptomatic of highly altered rivers, and low trophic diversity can make aquatic food webs less resilient in the face of further disturbance (Olden et al., 2004).

In summary, we found low diversity of benthic macroinvertebrate taxa at salinized sites in the Permian Basin region of the Pecos River in Texas (Orla and Iraan sites), in agreement with Davis (1980), but our results indicate more-recent declines in diversity further south in the transition to the Edwards Plateau (Sheffield site). As predicted, salt-tolerant taxa dominated assemblages, and groups considered particularly sensitive to salinization (Ephemeroptera, Plecoptera, and Trichoptera) were absent except for two tolerant trichopteran families. Results of this work add to our understanding of how dryland river salinization affects biodiversity and aquatic food web structure. Such information is important for conservation and management because predictions for future flow conditions in the Pecos River suggest that discharge will continue to decrease with climate change (United States Bureau of Reclamation [USBR], 2021), further reducing opportunities for salt export.

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